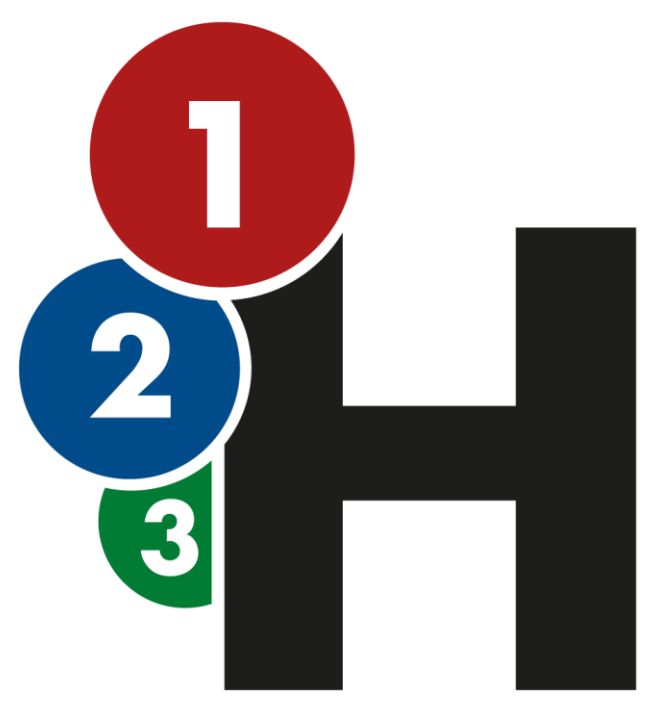




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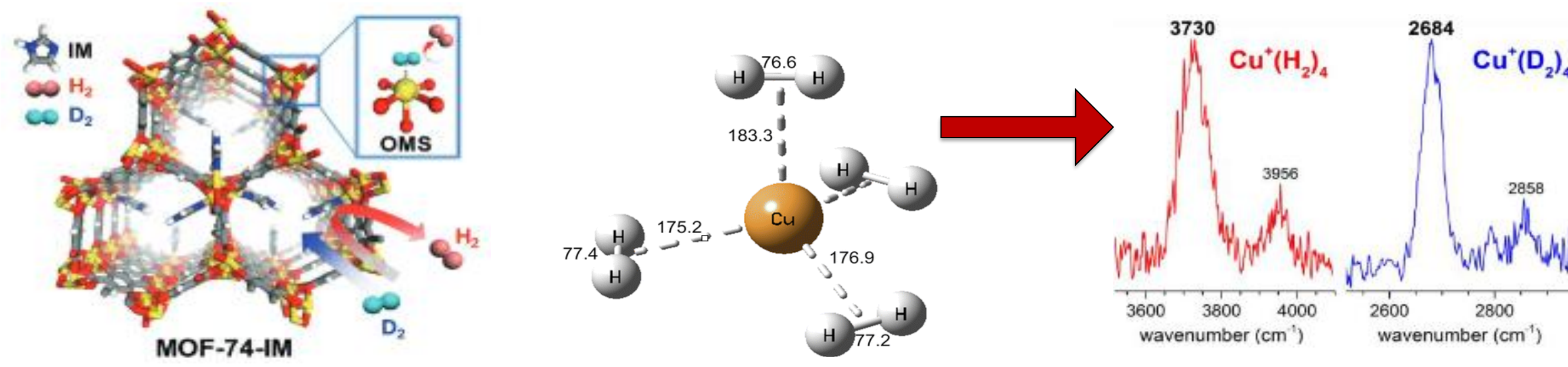
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Strong anharmonicity in the vibrational spectra of $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2/\text{D}_2)_n$ isotopologues

Abstract

- Understanding the model system $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_n$ and its bonding interaction promises great insights into hydrogen isotopologues separation to be used for designing porous materials such as MOFs and zeolites.^[1]



- According to vibrational second-order perturbation theory (VPT2),^[2] anharmonic contributions for the reaction $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2/\text{D}_2)_2 \rightarrow \text{Cu}^+(\text{H}_2\text{O})(\text{H}_2/\text{D}_2) + \text{H}_2/\text{D}_2$ add up to **0.76 kJ·mol⁻¹** / **0.37 kJ·mol⁻¹** which are higher compared to the obtained values **0.23 kJ·mol⁻¹** / **0.12 kJ·mol⁻¹** for the free H_2 and D_2 respectively.

- Excellent agreement between the theoretical and experimental spectra (cf. P1)

Methods

- Structure optimizations and all vibrational frequencies calculated at the **MP2/def2-TZVPP** and **CCSD(T)/aug-cc-pVTZ(-PP)** levels were carried out using the Gaussian 16 program package (for validation) and CFOUR 2.1.^[2]
- Energy calculations for the reaction $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_n \rightarrow \text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_{n-1} + \text{H}_2$ and its isotopologues were corrected with the use of anharmonic ZPEs.

Energetics

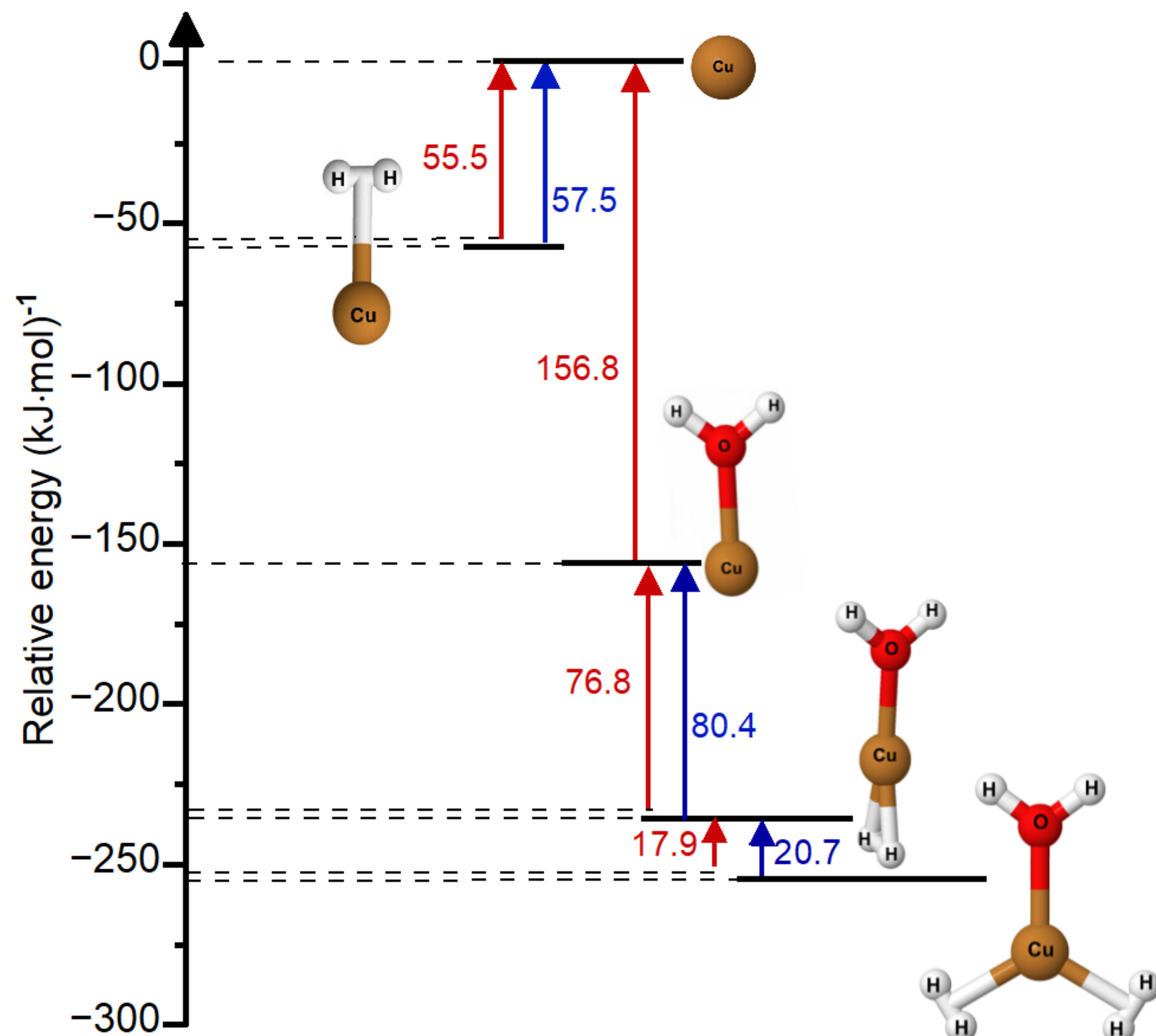


Fig. 1 : Relative energies (kJ·mol⁻¹) of $\text{Cu}^+(\text{H}_2/\text{D}_2)_{0.1}$ and $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2/\text{D}_2)_{0.1,2}$ complexes with anharmonic ZPE correction. All calculated at the MP2/def2-TZVPP level of theory. The red (values to the left of the arrow) represents the energy of the H_2 binding to Cu^+ while the blue (values to the right of the arrow) corresponds to the energy of D_2 binding to Cu^+ .

Summary

- Due to its lower zero-point energy (ZPE), D_2 preferentially binds to Cu^+ over H_2 by a value of 2.8 kJ·mol⁻¹.
- The shifts are observed in the spectra (Cf. Shabnam Haque Poster)

[1] Jin, J., Wulf, T., Jorewitz, M., Heine, T., & Asmis, K. R. Physical Chemistry., **2023**.

[2] Johannes Hoja and A. Daniel Boese, arXiv preprint arXiv:2209.04392., **2022**.

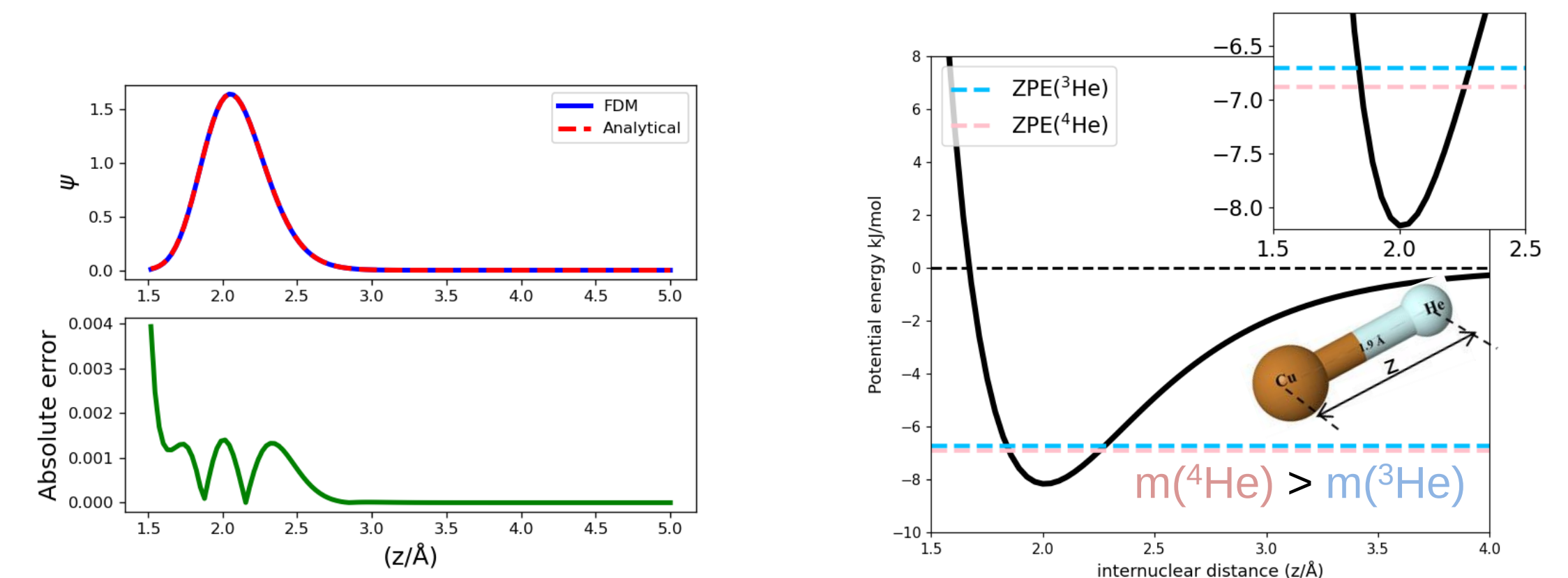
[3] Beenakker, J.; Borman, V.; Krylov, Chemical Physics Letters., **1995**

[4] Cho A. American Association for the Advancement of Science., **2009**

Computational modelling of atomistic motifs for adsorptive He isotope separation

Abstract

- Given the gap between the increasing helium demand and supply,^[4] we must seek a cost-effective method for helium isotope separation. Our project aims on the parametrisation of the **confining pore potential modeled numerically**.
- Finite difference method (FDM) confirmed for 1D SE in Cu^+ -He system at **CCSD(T)/aug-cc-pVTZ** level. It is a potential means to efficiently solve the 3D time independent Schrödinger equation (TISE).



- ^3He has a higher ZPE compared to ^4He due to its lighter nuclear mass.
- The selectivity (α) of He adsorption is dominated by the ZPE, $\alpha=2.6$ at $T=77\text{K}$

Methods and Results

- Ab initio calculations : PES, CCSD(T)/aug-cc-pVTZ, BSSE, CBS
- Analytical methods : Separation of variables and zeros of Bessels function
- Numerics : FDM with centrale difference scheme with three-point stencils

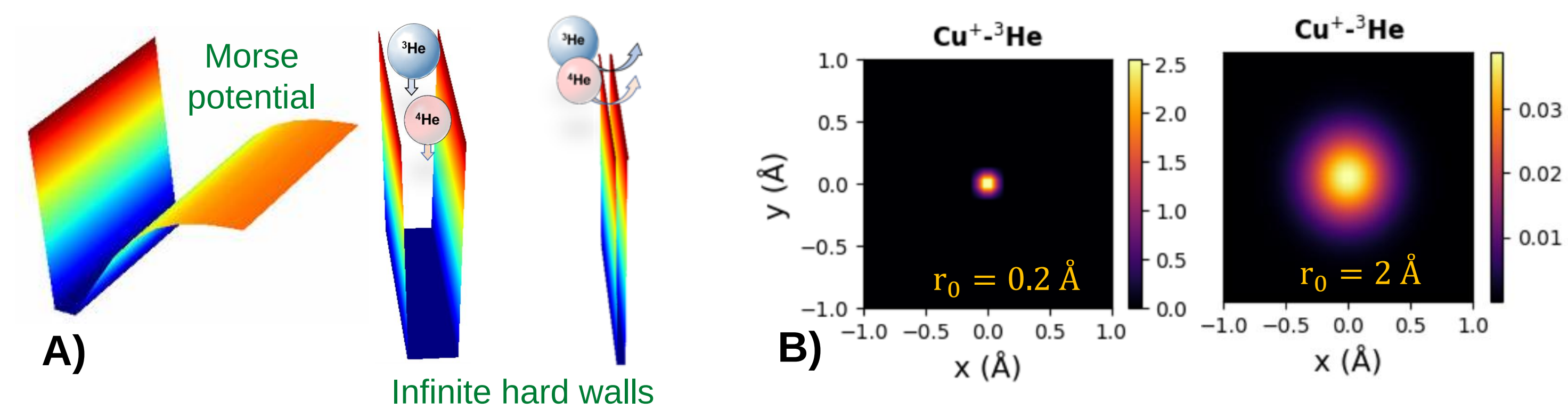
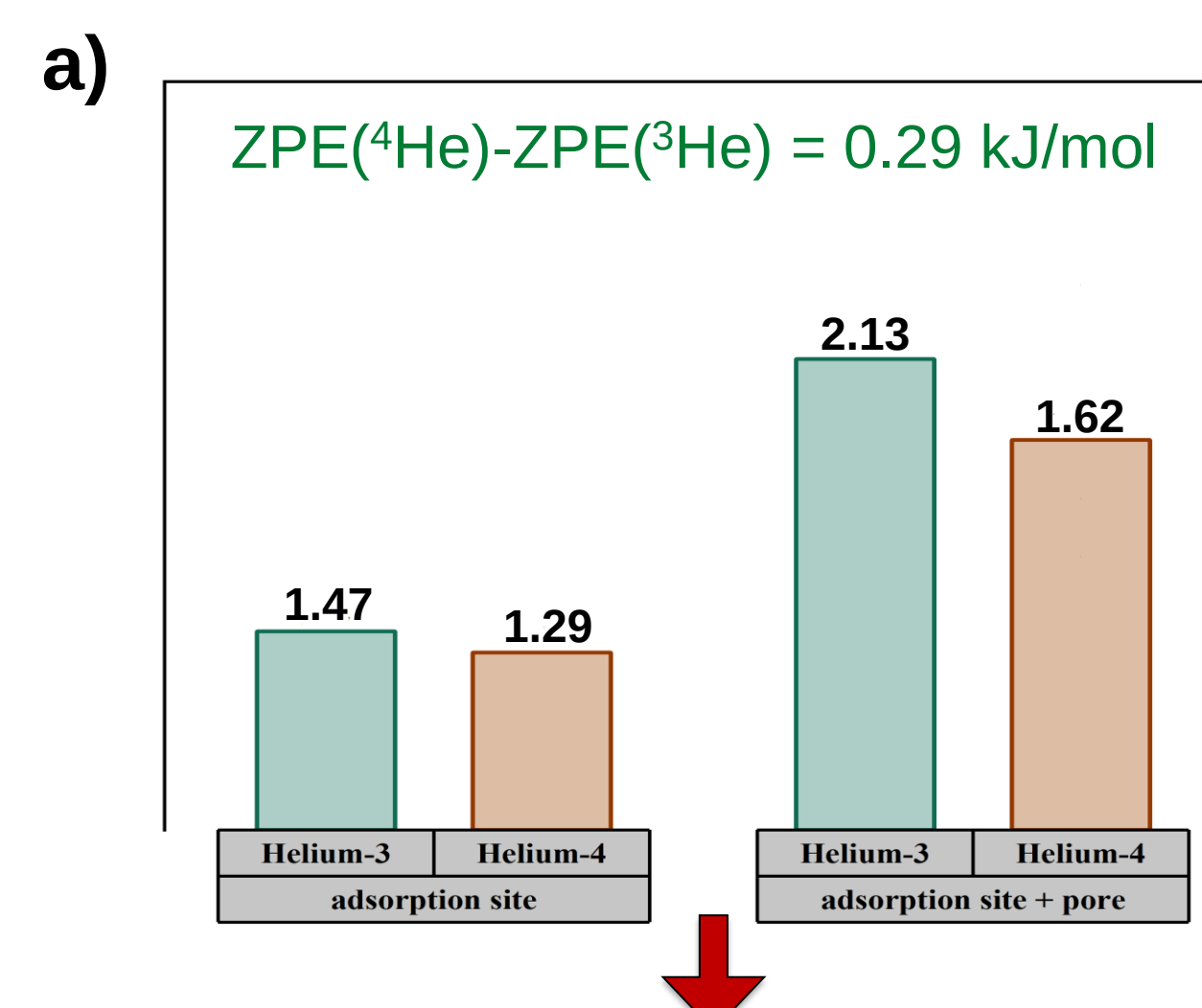


Fig. 2 : Plot of the probability density functions ($|\psi|^2$) for $\text{Cu(I)}\text{-He}$ system in presence of a of a hard walls potential. The box size is $1.5\text{\AA} \times 1.5\text{\AA} \times 1.5\text{\AA}$ (in x -, y -, and z -direction), with $N=42.875$ points.



Methods	ΔZPE kJ/mol	Time
Analytic	0.171	/
FDM ^w	0.172	0.05 s
VPT2	0.180	54 min
FDM ^c	0.291	23 min

^wZPE considering only the Morse potential. ^cZPE with presence of an infinite hard wall model ($r_0=0.1\text{ nm}$) in a 3D Scheme.

- The optimal hard wall radius lies in the range **[0.7-1] Å**. in particular **$0.8\text{\AA} \pm 0.02\text{\AA}$** .
- The Mathematical expression proposed by Beenakker et al.,^[3] failed because does not include any translational, rotational and vibrational partition functions.

Fig. 3 : **a)** ZPEs of helium isotopes within a strong adsorption site when confined within a hard wall model, coupled with strong adsorption site. **b)** $^4\text{He}/^3\text{He}$ isotopes selectivity for the adsorption of Helium on metal center at different temperature.

Outlook

- Theoretical investigation in CuPW-Alanine from Project 1
- Application of $^4\text{He}/^3\text{He}$ separation in MOFs
- FDM for hydrogen isotopologue separation in $\text{Cu}^+\text{-H}_2$

